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November 14, 2016

File Bridge Overview

Exemplar™

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File Bridge Overview

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Application Overview

The File Bridge at the highest level is a program that allows for other programs to access a file system that's present in an isolated network in a secure manor without needing inbound rules. There are three components to the file bridge in order to accomplish this. The Agent, Proxy, and the Client.

Proxy

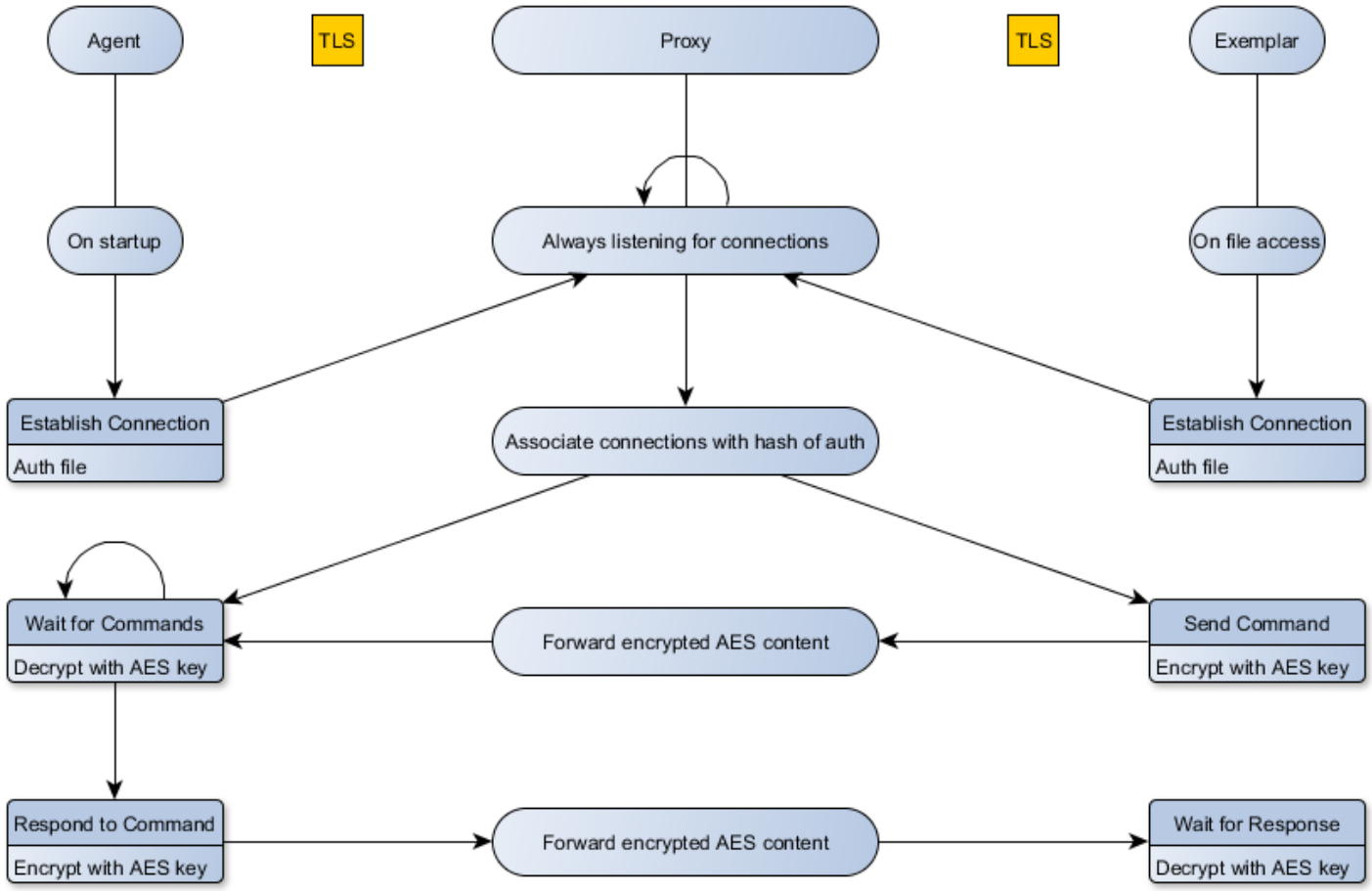
This program is essentially a proxy we host that the Repository Host and Client will use to communicate. By having the Proxy running between the Agent and Client we can have the Proxy listen for both sides of the connection so that inbound rules aren't required. Data between the Repository Host & Client is required to be encrypted, which the Proxy doesn't have the key for. So even if it wanted to it couldn't decrypt the data being routed through it.

Agent

This is the program that is installed in the isolated network and will connect to the Proxy via TLS and wait for operations to evaluate. The Agent will only run operations that are encrypted via the key defined in the initial connection configuration. After decrypting & evaluating an operation it will return the result of the operation with the same key so that only the Client can decrypt it.

Client

The client will commonly be an Exemplar instance running in the cloud, but the File Bridge API doesn't depend on Exemplar. Whenever the client needs to access the Agent it will make a connection with TLS to the Proxy and send an encrypted operation to be evaluated. It will then wait for a response and once it gets a response resume executing the logic that called the File Bridge API.

Diagram Below

Order of operations diagram